

1. Give a context-free grammar for each of the following languages.

a) $L = \{w \mid w \text{ contains even number of 0's}\}$

= $S \rightarrow 1S \mid 0T \mid \epsilon$

$T \rightarrow 0S \mid 1T$

b) $L = \{w \mid w \text{ contains twice as many 1s as 0s}\}$

= $S \rightarrow SS \mid A \mid B \mid C$

$A \rightarrow A011 \mid 0A11 \mid 01A1 \mid 011A \mid \epsilon$

$B \rightarrow B110 \mid 1B10 \mid 11B0 \mid 110B \mid \epsilon$

$C \rightarrow C101 \mid 1C01 \mid 10C1 \mid 101C \mid \epsilon$

c) $L = \{w \mid w \text{ contains even number of 0s and 1s}\}$

= $S \rightarrow 0X \mid 1Y \mid \epsilon$

$X \rightarrow 0S \mid 1Z$

$Y \rightarrow 1S \mid 0Z$

$Z \rightarrow 0Y \mid 1X$

d) $L = \{w \mid \text{where each 0's is followed by at least as many 1's}\}$

= $S \rightarrow AS \mid \epsilon$

$A \rightarrow 0A1 \mid 1A \mid \epsilon$

e) $L(G) = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i=j \text{ or } i=k\}. \Sigma = \{a, b, c\}$

= $S \rightarrow AC \mid S'$

$A \rightarrow aAb \mid \epsilon$

$C \rightarrow cC \mid \epsilon$

$S' \rightarrow aBc \mid B$

$B \rightarrow bB \mid \epsilon$

f) $L(G) = \{a^i b^j c^k \mid j > i+k\}. \Sigma = \{a, b, c\}$

= $S \rightarrow ABC$

$A \rightarrow aAb \mid \epsilon$

$B \rightarrow bB \mid b$

$C \rightarrow bCc \mid \epsilon$

$$g) L(G) = \{ a^n b^m \mid 0 < n < m < 3n \}. \Sigma = \{a, b\}$$

$$= \begin{array}{l} S \rightarrow aSbb \mid aSbbb \mid Zb \\ Z \rightarrow aZb \mid ab \end{array}$$

$$h) L(G) = \text{set of all strings } w \text{ over } \{a, b\} \text{ such that } w \text{ is not palindrome.}$$

$$= \begin{array}{l} Y \rightarrow aYa \mid bYb \mid aZb \mid bZa \\ Z \rightarrow aZ \mid bZ \mid \epsilon \end{array}$$

$$i) L = \{w \mid w = w^R \text{ AND } |w| \text{ is even, } w \text{ is a palindrome}\}$$

$$= \begin{array}{l} S \rightarrow A0A \mid B1B \mid \epsilon \\ A \rightarrow 1A \mid 0A \mid \epsilon \\ B \rightarrow 1B \mid 0B \mid \epsilon \end{array}$$

$$j) L(G) = \{ a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i=j \text{ or } j=k \}. \Sigma = \{a, b, c\}$$

$$= \begin{array}{l} S \rightarrow AC \mid S' \\ A \rightarrow aAb \mid C \\ C \rightarrow cC \mid \epsilon \\ S' \rightarrow A'B \\ A' \rightarrow aA' \mid \epsilon \\ B \rightarrow bBb \mid A' \end{array}$$

$$k) L(G) = \{ a^n b^m c^m d^{2n} \mid n \geq 0, m > 0 \}$$

$$= \begin{array}{l} S \rightarrow aBdd \mid A \\ A \rightarrow aSdd \mid \epsilon \\ B \rightarrow bBc \mid bc \end{array}$$

$$l) L = \{w \mid w \text{ contains at least 4 a's}\}$$

$$= \begin{array}{l} S \rightarrow RaRaRaRaR \\ R \rightarrow bR \mid aR \mid \epsilon \end{array}$$

2. What does the following CFGs do?

$$a) \begin{array}{l} S \rightarrow ZSZ \mid 0 \\ Z \rightarrow 0 \mid 1 \end{array}$$

$$= L = \{w \mid \text{the length of } w \text{ is odd and its middle is } 0\}$$

$$b) \begin{array}{l} S \rightarrow 0E0 \mid 1E1 \mid \epsilon \\ E \rightarrow 1E \mid 0E \mid \epsilon \end{array}$$

$$= L = \{w \mid w \text{ starts and ends with the same symbol}\}$$

$$\begin{aligned} \text{c)} \quad & S \rightarrow AB \\ & A \rightarrow 0A1 \mid \epsilon \\ & B \rightarrow 1B \mid \epsilon \end{aligned}$$

$$= L(G) = \{0^m 1^{m+n} \mid n, m \geq 0\} \text{ over the terminals } \{0,1\}$$

$$\text{d)} \quad S \rightarrow \epsilon \mid 1S1S0S \mid 1S1S0S1S \mid 1S0S1S1S \mid 0S1S1S1S$$

$$= L = \{w \mid w \text{ contains thrice as many 1s as 0s}\}$$

$$\text{e)} \quad S \rightarrow aSbb \mid aSb \mid \epsilon$$

$$= L(G) = \{a^n b^m \mid 2n \geq m \geq n \geq 0\} \text{ over the terminals } \{0,1\}$$

3. Convert the following Regular expressions to a CFG.

$$\text{a)} \quad a(b \mid c^*)$$

$$\begin{aligned} = \quad & S \rightarrow aX \\ & X \rightarrow b \mid C \\ & C \rightarrow Cc \mid \epsilon \end{aligned}$$

$$\text{b)} \quad 0^*1(0 + 1)^*$$

$$\begin{aligned} = \quad & S \rightarrow A1B \\ & A \rightarrow 0A \mid \epsilon \\ & B \rightarrow 0B \mid 1B \mid \epsilon \end{aligned}$$

$$\text{c)} \quad (a + b)^*(a^* + (ba)^*)$$

$$\begin{aligned} = \quad & V \rightarrow WX \\ & W \rightarrow aW \\ & W \rightarrow bW \\ & W \rightarrow \epsilon \\ & X \rightarrow Y \\ & X \rightarrow Z \\ & Y \rightarrow aY \\ & Y \rightarrow \epsilon \\ & Z \rightarrow baZ \end{aligned}$$

$$Z \rightarrow \epsilon$$

d) $(a+b)^* aa (a+b)^*$

= $S \rightarrow AaaA$

$$A \rightarrow aA \mid bA \mid \epsilon$$

e) $a^* + a(a|b)^*$

= $S \rightarrow X|Y$

$$X \rightarrow aX \mid \epsilon$$

$$Y \rightarrow aZ$$

$$Z \rightarrow aZ \mid bZ \mid \epsilon$$

4. Consider the following context-free grammar $\Sigma=\{0,1\}$. Give leftmost and rightmost derivations for the following strings and check parse-tree ambiguity.

a) $S \rightarrow 0A \mid 1B$

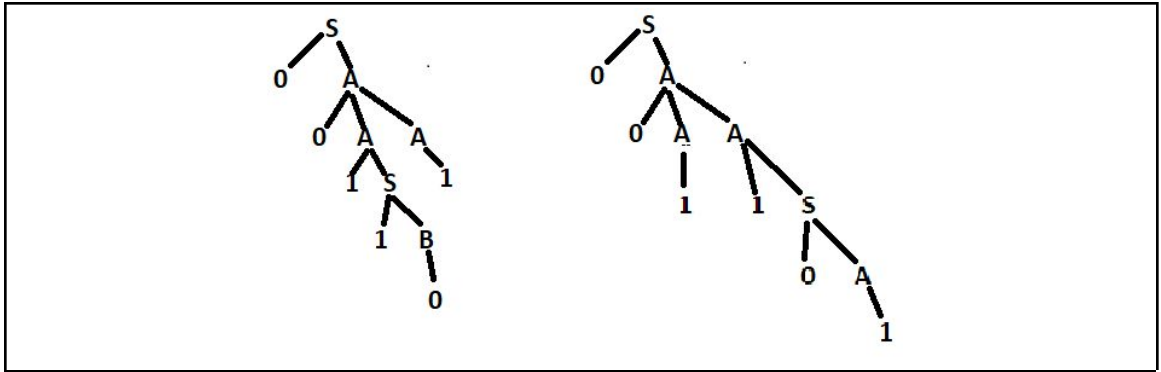
$$A \rightarrow 0AA \mid 1S \mid 1$$

$$B \rightarrow 0S \mid 1BB \mid 0$$

Strings: 001101

leftmost derivation:	rightmost derivation:
$S \rightarrow 0A$	$S \rightarrow 0A$
$\rightarrow 00AA$	$\rightarrow 00AA$
$\rightarrow 001A$	$\rightarrow 00A1$
$\rightarrow 0011S$	$\rightarrow 001S1$
$\rightarrow 00110A$	$\rightarrow 0011B1$
$\rightarrow 001101$	$\rightarrow 001101$

we can find two parse trees for this grammar, so the grammar is ambiguous.



- b) $S \rightarrow A 1 B$
 $A \rightarrow 0A \mid \epsilon$
 $B \rightarrow 0B \mid 1B \mid \epsilon$
- Strings: 10100, 0010101
- = for string 10100:

leftmost derivation:	rightmost derivation:	Parse Tree:
$S \rightarrow A1B$ $\rightarrow \epsilon 1B$ $\rightarrow 10B$ $\rightarrow 101B$ $\rightarrow 1010B$ $\rightarrow 10100\epsilon$ $\rightarrow 10100$	$S \rightarrow A1B$ $\rightarrow A10B$ $\rightarrow A101B$ $\rightarrow A1010B$ $\rightarrow A10100$ $\rightarrow \epsilon 10100$ $\rightarrow 10100$	

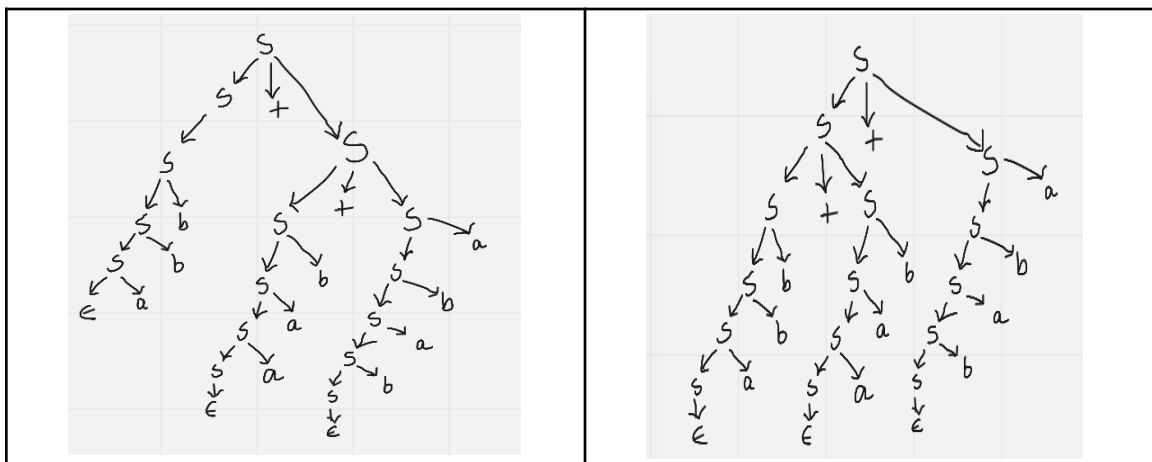
for string 0010101:

leftmost derivation:	rightmost derivation:	Parse Tree:
$S \rightarrow A1B$ $\rightarrow 0A1B$ $\rightarrow 00A1B$ $\rightarrow 00\epsilon 1B$ $\rightarrow 001B$ $\rightarrow 0010B$	$S \rightarrow A1B$ $\rightarrow A10B$ $\rightarrow A101B$ $\rightarrow A101\epsilon$ $\rightarrow 0A101$ $\rightarrow 00A101$	

- d) $S \rightarrow S + S$
 $S \rightarrow Sa \mid Sb \mid \epsilon$
 String: abb + aab + baba

Leftmost Derivation:	Rightmost Derivation:
$S \rightarrow S + S$	$S \rightarrow S + S$
$\rightarrow Sb + S$	$\rightarrow S + Sa$
$\rightarrow Sbb + S$	$\rightarrow S + Sba$
$\rightarrow Sabb + S$	$\rightarrow S + Saba$
$\rightarrow abb + S + S$	$\rightarrow S + Sbaba$
$\rightarrow abb + Sb + S$	$\rightarrow S + baba$
$\rightarrow abb + Sab + S$	$\rightarrow S + S + baba$
$\rightarrow abb + Saab + S$	$\rightarrow S + Sb + baba$
$\rightarrow abb + aab + Sa$	$\rightarrow S + Sab + baba$
$\rightarrow abb + aab + Sba$	$\rightarrow S + Saab + baba$
$\rightarrow abb + aab + Saba$	$\rightarrow S + aab + baba$
$\rightarrow abb + aab + Sbaba$	$\rightarrow Sb + aab + baba$
$\rightarrow abb + aab + baba$	$\rightarrow Sbb + aab + baba$
	$\rightarrow Sabb + aab + baba$
	$\rightarrow abb + aab + baba$

we can find two parse trees for this grammar, so the grammar is ambiguous.



- e) $S \rightarrow SA \mid \epsilon$
 $A \rightarrow aa \mid ab \mid ba \mid bb$
 String: aabbbba

Leftmost Derivation: $S \rightarrow SA$ $\rightarrow SAA$ $\rightarrow SAAA$ $\rightarrow aaAA$ $\rightarrow aabbA$ $\rightarrow aabbba$	Rightmost Derivation: $S \rightarrow SA$ $\rightarrow Sba$ $\rightarrow SAba$ $\rightarrow Sbbba$ $\rightarrow SAbbba$ $\rightarrow Saabbba$ $\rightarrow aabbba$	
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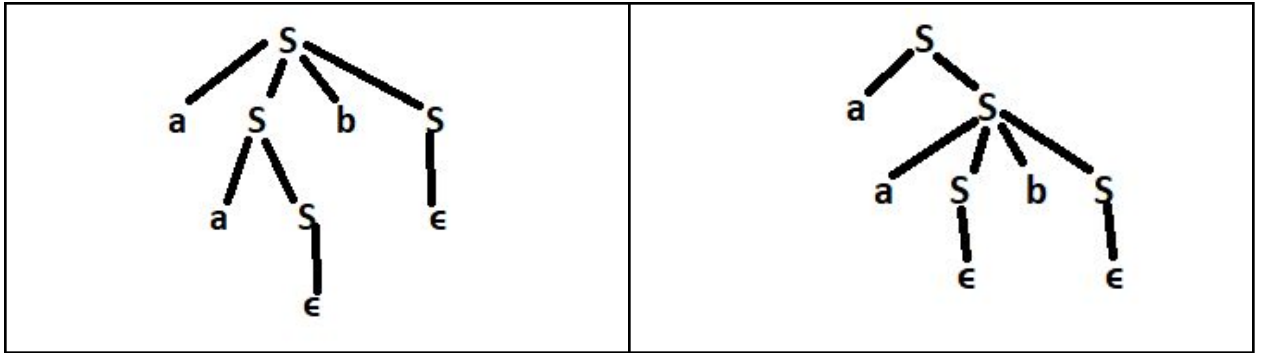
The grammar is unambiguous since only one parse tree is possible for every string.

- f)
- $S \rightarrow aEbS$
- $S \rightarrow aEbScS \mid \epsilon$
- $E \rightarrow d$

String: adbadbadbc

Leftmost Derivation: $S \rightarrow aEbS$ $\rightarrow adbS$ $\rightarrow adbaEbS$ $\rightarrow adbadbS$ $\rightarrow adbadbaEbScS$ $\rightarrow adbadbadbScS$ $\rightarrow adbadbadbc$	Rightmost Derivation: $S \rightarrow aEbS$ $\rightarrow aEbaEbScS$ $\rightarrow aEbaEbSc\epsilon$ $\rightarrow aEbaEbSc$ $\rightarrow aEbaEbaEbSc$ $\rightarrow aEbaEbaEb\epsilon c$ $\rightarrow aEbadbadbc$ $\rightarrow adbadbadbc$
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we can find two parse trees for this grammar, so the grammar is ambiguous.



The grammar is ambiguous but not inherently ambiguous. So, equivalent unambiguous CFG:

$S \rightarrow aS \mid aTbS \mid \epsilon$

$T \rightarrow aTbT \mid \epsilon$